IGCSE · Cambridge (CIE) · Maths

59 mins   22 questions

Calculator Questions

Expanding & Factorising Brackets

Expanding & Simplifying Single Brackets / Expanding Double Brackets / Expanding Triple Brackets / Factorising Out Terms / Factorising by Grouping / Factorising Simple Quadratics / Factorising Harder Quadratics / Difference of Two Squares / Deciding the Factorisation Method

Easy (3 questions)	/6
Medium (6 questions)	/14
Hard (7 questions)	/19
Very Hard (6 questions)	/20
Total Marks	**/59**

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Easy Questions

1 Factorise completely.

$$4 - 8x$$

(2 marks)

2 Factorise completely.

$$3x^2 - 12xy$$

(2 marks)

3 Expand and simplify $(x+7)(x-3)$.

(2 marks)

Medium Questions

1 (a) Factorise.

$xy + 2wx$

(1 mark)

(b) $2px - x + 14p - 7$

(2 marks)

2 Factorise completely.

$$3a^2b - ab^2$$

(2 marks)

3 Factorise completely.

i) $6y^2 - 15xy$

[2]

ii) $y^2 - 9x^2$

[2]

(4 marks)

4 Factorise $x^2 - 25$.

(1 mark)

5 Expand and simplify $5(p+3) - 2(1-2p)$

(2 marks)

6 Expand and simplify $(p+9)(p-4)$

(2 marks)

Hard Questions

1 Expand and simplify.

$(x+7y)(x-y)(x-6y)$

(3 marks)

2 Factorise

$2x + 6 - 3xy - 9y$

(2 marks)

3 Factorise $24 + 5x - x^2$.

(2 marks)

4 Expand and simplify $(3x-5y)(2x+y)$.

(2 marks)

5 Factorise $5m^2 - 20p^4$.

(3 marks)

6 Factorise $12ab^3 + 18a^3b^2$.

(2 marks)

7 Factorise.

i) $2mn + m^2 - 6n - 3m$

[2]

ii) $4y^2 - 81$

[1]

iii) $t^2 - 6t + 8$

[2]

(5 marks)

Very Hard Questions

1 Expand and simplify.

$$(x-2)(x+5)(2x-1)$$

(3 marks)

2 Write $(x + 1)(x - 3)^2$ in the form $ax^3 + bx^2 + cx + d$.

(3 marks)

3 Expand and simplify.

$$(y+3)(y-4)(2y-1)$$

(3 marks)

4 Factorise completely.

i) $15p^2q^2 - 25q^3$

[2]

ii) $4fg + 6gh + 10fk + 15hk$

[2]

iii) $81k^2 - m^2$

[2]

(6 marks)

5 Factorise $3x^2 + 11x - 20$

(2 marks)

6 Factorise completely.

$$2t^2 - 98m^2$$

(3 marks)